

More on conditional statements:-

$$\Rightarrow \underbrace{(x \text{ is a multiple of } 6)}_P \Rightarrow \underbrace{(x \text{ is even})}_Q$$

True

$$\forall x \in \mathbb{Z}, P \Rightarrow Q$$

$$\Rightarrow \underbrace{(x \text{ is even})}_Q \Rightarrow \underbrace{(x \text{ is a multiple of } 6)}_P$$

open sentence

$$\Rightarrow \forall x \in \mathbb{Z}, Q \Rightarrow P = \text{false}$$

$$\forall x \in \mathbb{Z}, (P(x) \Rightarrow Q(x))$$

eg:- $(x \in \mathbb{R}) \Rightarrow (x^2 + 1 > 0)$ true

$$\forall n \in \mathbb{N}, (n \text{ is a prime number}) \Rightarrow (n \text{ is odd})$$

$$n = 2$$